

Pooling Hides It: On Evaluating Sampling-Based Uncertainty for Vision–Language Models on Handwritten Mathematics

Abstract

Sampling a vision–language model repeatedly and measuring how much it disagrees with itself is a cheap, label-free uncertainty signal. We ask what that signal actually measures on handwritten mathematics, and find that the answer depends as much on how it is evaluated as on the model. On transcription, self-disagreement predicts error well (AUROC 0.835), survives every artifact control we apply, and beats the model’s own token confidence by a resolved margin; a rule that abstains whenever all five samples disagree flags a genuinely wrong transcription 57 times out of 61. On grading, the pooled statistic reads 0.520 — indistinguishable from chance — and we show this is an evaluation artifact rather than a null result: the model answers “there is an error” for most inputs, and uncertainty predicts its mistakes in *opposite directions* on erroneous (0.801) and correct (0.280) solutions, so pooling cancels a real effect. The stratified effect replicates across three independently trained model families. We then report two ways the evaluation itself misleads: a fourth model attains the highest raw score in our study (0.915) purely by generating incoherent output on two-thirds of inputs, and a second dataset cannot test the effect at all because it contains no correct solutions. **[TODO: tighten to the venue’s word limit once the body is final]**

1 Introduction

[TODO: write. Do not open with “Large language models have recently...”]

2 Related Work

[TODO: write. VERIFY EVERY CITATION AGAINST THE REAL PAPER – do not cite from memory. This section is the largest genuinely-new writing task.]

3 Setup

Data. FERMAT [1] contains $\sim 2,200$ photographed handwritten solutions to middle- and high-school problems, in which errors were deliberately introduced by the dataset authors and annotated by type. Each image contains a complete question and its answer, and 15% of items are error-free. That last property is load-bearing for this paper: §4.6 shows the

analysis in §4.4 is impossible on datasets that lack it. Unless stated otherwise we use a balanced $n = 300$ sample (150 with an error, 150 without), drawn once with a fixed seed.

Tasks. We evaluate two distinct tasks on the same images. *Perception* asks the model to “explicitly perform OCR on the handwritten text and extract the content in $\text{\LaTeX}$ format”; it is correct when the extracted final answer matches the reference after canonicalisation. *Reasoning* asks it to “analyze the Answer to determine whether there is any error” and emit a binary judgement; it is correct when that judgement matches the annotation. The model is never shown the reference answer.

Uncertainty measure. For each item we draw $K = 5$ samples at temperature 0.7, reduce each to a canonical label, and take the Shannon entropy of the resulting cluster distribution, $H = -\sum_i p_i \ln p_i$. $H = 0$ when all samples agree and $H = \ln K$ when all differ. We deliberately use exact-match clustering over an *extracted answer span* rather than semantic clustering of whole derivations; §6 records why.

Models. Qwen2.5-VL at 3B and 7B, LLaVA-NeXT-7B, and InternVL3-8B — three independently trained families, all run in bfloat16.

Statistics. Every AUROC carries a 95% bootstrap interval over 10,000 item-level resamples. Comparisons between conditions use a *paired* bootstrap over the same resampled items rather than checking whether two marginal intervals overlap. Before any run we registered a minimum minority-class size of 30 and, for the stratified reasoning hypothesis, an AUROC threshold of 0.70; verdicts below are assigned by a pure function of those thresholds, not by inspection.

4 Results

4.1 Does self-disagreement predict transcription error?

Yes, and by a margin that does not depend on the model’s own confidence being poor. On the balanced $n = 300$ sample, perception entropy predicts transcription error with AUROC 0.835 [0.787, 0.879]. The model’s own mean token log-probability reaches only 0.537 [0.469, 0.605] — an interval that includes chance — and the paired difference is +0.297

Signal	AUROC (95% CI)	vs. entropy	Subset	n	AUROC (95% CI)
Sampling entropy	0.835 [0.787, 0.879]	—	All items	300	0.835 [0.787, 0.879]
1—majority fraction	0.793 [0.742, 0.841]	+0.042 [+0.011, +0.073]	Excl. parse failures	255	0.839 [0.790, 0.886]
Distinct count	0.790 [0.739, 0.838]	+0.045 [+0.016, +0.075]	Excl. max-entropy items	239	0.794 [0.737, 0.848]
Majority margin	0.779 [0.727, 0.830]	+0.055 [+0.018, +0.092]	Excl. both	206	0.796 [0.736, 0.852]
Token log-probability	0.537 [0.469, 0.605]	+0.297 [+0.214, +0.380]			

Table 1: Perception baselines, Qwen2.5-VL-3B, $n = 300$. The upper block reduces the same five samples in different ways; the lower is a model-internal signal requiring no sampling. Differences are paired bootstraps over the same resampled items, and every one excludes zero. Entropy’s margin over the simpler summaries is small but resolved, so the shape of the cluster distribution carries information that a count or a majority does not. The simple summaries are related to entropy but not redundant with it (Spearman 0.83–0.89).

[+0.214, +0.380], resolved. The signal is therefore not recoverable from information the model already exposes.

Read as a deferral rule the result is directly usable: of the 61 items on which all five transcriptions disagree, 57 are wrong (93.4% precision, 31.3% recall). Over the full risk–coverage curve this gives AUC 0.410 against a 0.607 random-deferral baseline. **[TODO: Fig. 1: risk–coverage curve. Asset exists at report/figures/risk.coverage.n300.png]**

4.2 What does the entropy summary add?

All five samples per item support several summaries, and a natural objection is that entropy is merely a dressed-up count of distinct answers. Table 1 tests that directly. The three sample-based alternatives are computed from the *same* generations at no additional cost, so the comparison isolates the summary rather than the sampling budget.

Two of these deserve comment. The majority margin separates a three–two split from a three–one–one split, which the majority fraction cannot, and it is nonetheless the weakest of the three — the information entropy uses is in the full distribution, not in the top two clusters. Token log-probability is the only baseline here that is not a function of the samples, and it is also the only one that fails to beat chance; a model that is verbally or numerically confident about a page it has misread is not detectable from its own output distribution, but is detectable from asking it twice. **[TODO: we also have verbalized confidence, but only on the grading arm (0.547 [0.432, 0.661], no signal, $n=100$). Either run it for perception or report it in §4.4 where the data exists — do not quietly omit it.]**

4.3 Does the signal survive artifact controls?

A cluster-entropy AUROC can be manufactured two ways that have nothing to do with useful uncertainty. Unparseable samples receive their own cluster label, which both inflates entropy and correlates with being scored wrong, so the metric could be measuring the parser. And items pinned at the ceiling

Table 2: Perception entropy under artifact controls, Qwen2.5-VL-3B. The signal is graded rather than concentrated at the ceiling: it survives the harshest simultaneous cut with an interval excluding chance. §4.7 shows a model for which it does not.

Scored on	n_{wrong}	AUROC (95% CI)	Verdict
Pooled	123	0.520 [0.458, 0.582]	no signal
Erroneous solutions	73	0.801 [0.751, 0.846]	confirmed
Correct solutions	106	0.280 [0.200, 0.366]	inverted

Table 3: Qwen2.5-VL-7B. Both strata are adequately powered and each excludes chance; the pooled statistic reports neither. Note the inversion is a finding, not an absence: entropy ranks correct solutions *backwards* because the model is confidently wrong on them.

$H = \ln K$ can carry the entire ranking if that cell happens to be uniformly incorrect, leaving a binary breakdown detector rather than a graded signal. Table 2 removes each, and both.

4.4 Why does grading look like chance?

Pooled over the same 300 items, reasoning entropy predicts grading error with AUROC 0.520 [0.458, 0.582] — no signal. Read alone, that is a clean negative result, and we initially reported it as one.

It is an artifact of aggregation. The model answers “there is an error” for the large majority of inputs, so its mistakes are of opposite kinds on the two halves of the data: on solutions that do contain an error it is usually right and errs by missing one, while on correct solutions it is usually wrong and errs by inventing one. Entropy tracks both, in opposite directions, and pooling cancels them (Table 3).

Two consequences are worth stating separately from the numbers. First, the pooled statistic is not merely uninformative here, it is misleading: on a sample that is not balanced it drifts toward whichever stratum is larger, which can make an unchanged effect appear to strengthen. **[TODO: cite our own $n=500/n=800$ runs where pooled reads ~ 0.61 purely from the 650/150 composition]** Second, the diagnosis is only available because the dataset contains correct solutions to separate out.

4.5 Does the stratified effect replicate?

Across three independently trained families, yes, at magnitudes whose intervals overlap (Table 4). This is measured on matched item sets where possible: the two Qwen results use essentially the same 650 items.

Model	n	n_{wrong}	AUROC (95% CI)
Qwen2.5-VL-3B	650	42	0.854 [0.796, 0.902]
Qwen2.5-VL-7B	648	73	0.801 [0.751, 0.846]
LLaVA-NeXT-7B	400	34	0.775 [0.694, 0.848]

Table 4: Reasoning entropy on erroneous solutions. Each clears the pre-registered 0.70 threshold and the $n=30$ minority-class minimum.

4.6 What happens on a second dataset?

The analysis in §4.4 requires a dataset containing correct solutions, and the two public alternatives we examined do not have any: in ErrorRadar [2] no item is annotated error-free, and in ScratchMath [3] the student answer never equals the reference. Neither can be balanced, and neither can exhibit the inversion.

On ScratchMath, which is the only one of the two with handwriting to grade, the measurement fails for a second and more instructive reason. Grading accuracy reads 96.0%, but every item contains an error and the model answers “error” for 90% of them, so it scores well by construction. In 24% of readings (120/500) the model explicitly states it cannot read the image, and in 82% of those (98/120) it answers “error” regardless. Excluding those readings does not clean the estimate: all four misgraded items lie among them, so the retained subset has no minority class and the AUROC is undefined rather than improved. We therefore report this setting as *gated*: not a negative result about uncertainty, but a statement that the pair cannot test it.

4.7 When does a high score mean nothing?

InternVL3-8B attains the highest raw perception AUROC in our study, 0.915 [0.880, 0.944], on a transcription accuracy of 21.0% — about half that of the model in §4.1. Independently recomputing every derived column from raw output reproduces the stored values exactly, so this is model behaviour rather than a scoring defect. Under the control of §4.3 it collapses to 0.556 [0.495, 0.614].

The mechanism is visible in the entropy distribution: 202 of 300 items sit at the ceiling, and accuracy within that group is 0.99% against 62.2% elsewhere. The score is separating total generation breakdown from everything else, which is a binary property of the output, not graded uncertainty about content. Reported without the control, it would have been the strongest number in this paper. **[TODO: Fig. 2: the four-quadrant figure (low/high entropy $\times$ correct/wrong) with real pages. Assets exist at reference/cases/01–04.]**

5 Failure Analysis

[TODO: expand to the 50–100 coded cases. Tooling exists (pilot/07_manual_case_inspection.ipynb); this is reading time, not compute. Report inter-rater agreement if a second reader is available.]

6 Limitations

Every quantitative claim rests on a single dataset. We attempted a second and report in §4.6 why it could not test the central result — a property of the available benchmarks rather than of our design, but a limitation nonetheless.

The perception result is confirmed on one model family only. Of the two alternatives we tried, one is at a floor on free-response transcription (0.8% accuracy) and cannot test it, and the other fails the artifact control of §4.3. Whether perception entropy generalises across families is therefore open.

Our clustering is exact-match over an extracted answer span, not semantic. Semantic clustering was attempted and abandoned: on this data an NLI-plus-union-find implementation merges opposed clusters through a single false-positive transitive link, and degrades sharply at larger K . The reported results avoid that path entirely rather than depending on it being fixed.

Confidence intervals were added after the experiments were designed and run. They corrected several conclusions but cannot supply statistical power the studies were not sized for. **[TODO: add: $K=5$ gives few operating points, which is why two-thirds of conformal calibration splits cannot reach a 30% risk target.]**

7 Conclusion

[TODO: write, ~150 words. The line to land: whether sampling-based uncertainty “works” is not a property of the method alone — on the same images and the same measure it is genuinely useful, apparently absent, and spuriously excellent, depending on how the evaluation is set up.]

References

- [1] “Can vision-language models evaluate handwritten math?,” *arXiv preprint arXiv:2501.07244*, 2025. VERIFY authors and venue.
- [2] “Errorradar: Benchmarking complex mathematical reasoning of multimodal large language models via error detection,” *arXiv preprint arXiv:2410.04509*, 2024. VERIFY authors and venue.
- [3] “Scratchmath.” VERIFY – dataset used via HuggingFace songdj/ScratchMath; find and cite the accompanying paper.